

RW3 Separate Sheet Module - CME

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Input workbook: full_uah.xlsx

Source sheet: CME III

Rows (clean): 1023

Date range: 2022-01-03 .. 2026-01-27

Products: 1

Stats rows: 3

Tests rows: 3

Interpretation option: use tests table -> recommended_action/recommended_model_family per product.

Table: clean

source	date	raw_product	product	standardized_type	brand	region	price	unit_ok
CME	2026-01-27	CME Class III	Молоко	питне	milk	US	633.94191	1
CME	2026-01-26	CME Class III	Молоко	питне	milk	US	635.007552	1
CME	2026-01-23	CME Class III	Молоко	питне	milk	US	634.1188229999999	1
CME	2026-01-22	CME Class III	Молоко	питне	milk	US	635.981007	1
CME	2026-01-21	CME Class III	Молоко	питне	milk	US	637.5374280000001	1
CME	2026-01-20	CME Class III	Молоко	питне	milk	US	638.20005	1
CME	2026-01-16	CME Class III	Молоко	питне	milk	US	639.608398	1
CME	2026-01-15	CME Class III	Молоко	питне	milk	US	635.641864	1
CME	2026-01-14	CME Class III	Молоко	питне	milk	US	637.712474	1
CME	2026-01-13	CME Class III	Молоко	питне	milk	US	638.879304	1
CME	2026-01-12	CME Class III	Молоко	питне	milk	US	637.52036	1
CME	2026-01-09	CME Class III	Молоко	питне	milk	US	638.837344	1
CME	2026-01-08	CME Class III	Молоко	питне	milk	US	636.033795	1
CME	2026-01-07	CME Class III	Молоко	питне	milk	US	641.856072	1
CME	2026-01-06	CME Class III	Молоко	питне	milk	US	640.129872	1
CME	2026-01-05	CME Class III	Молоко	питне	milk	US	635.6818259999999	1
CME	2026-01-02	CME Class III	Молоко	питне	milk	US	635.5034069999999	1
CME	2025-12-31	CME Class III	Молоко	питне	milk	US	645.142316	1
CME	2025-12-30	CME Class III	Молоко	питне	milk	US	670.420252	1
CME	2025-12-29	CME Class III	Молоко	питне	milk	US	667.5509189999999	1
CME	2025-12-26	CME Class III	Молоко	питне	milk	US	665.058966	1
CME	2025-12-24	CME Class III	Молоко	питне	milk	US	663.9217309999999	1
CME	2025-12-23	CME Class III	Молоко	питне	milk	US	665.098596	1
CME	2025-12-22	CME Class III	Молоко	питне	milk	US	666.252537	1
CME	2025-12-19	CME Class III	Молоко	питне	milk	US	667.660798	1
CME	2025-12-18	CME Class III	Молоко	питне	milk	US	668.523336	1
CME	2025-12-17	CME Class III	Молоко	питне	milk	US	667.785384	1
CME	2025-12-16	CME Class III	Молоко	питне	milk	US	669.162384	1
CME	2025-12-15	CME Class III	Молоко	питне	milk	US	667.080816	1
CME	2025-12-12	CME Class III	Молоко	питне	milk	US	671.7036690000001	1
CME	2025-12-11	CME Class III	Молоко	питне	milk	US	671.0026439999999	1
CME	2025-12-10	CME Class III	Молоко	питне	milk	US	673.253448	1
CME	2025-12-09	CME Class III	Молоко	питне	milk	US	671.8579000000001	1
CME	2025-12-08	CME Class III	Молоко	питне	milk	US	670.383798	1
CME	2025-12-05	CME Class III	Молоко	питне	milk	US	675.7844759999999	1
CME	2025-12-04	CME Class III	Молоко	питне	milk	US	679.4747400000001	1
CME	2025-12-03	CME Class III	Молоко	питне	milk	US	668.8803600000001	1
CME	2025-12-02	CME Class III	Молоко	питне	milk	US	726.5767079999999	1
CME	2025-12-01	CME Class III	Молоко	питне	milk	US	725.2845599999999	1
CME	2025-11-28	CME Class III	Молоко	питне	milk	US	725.71616	1

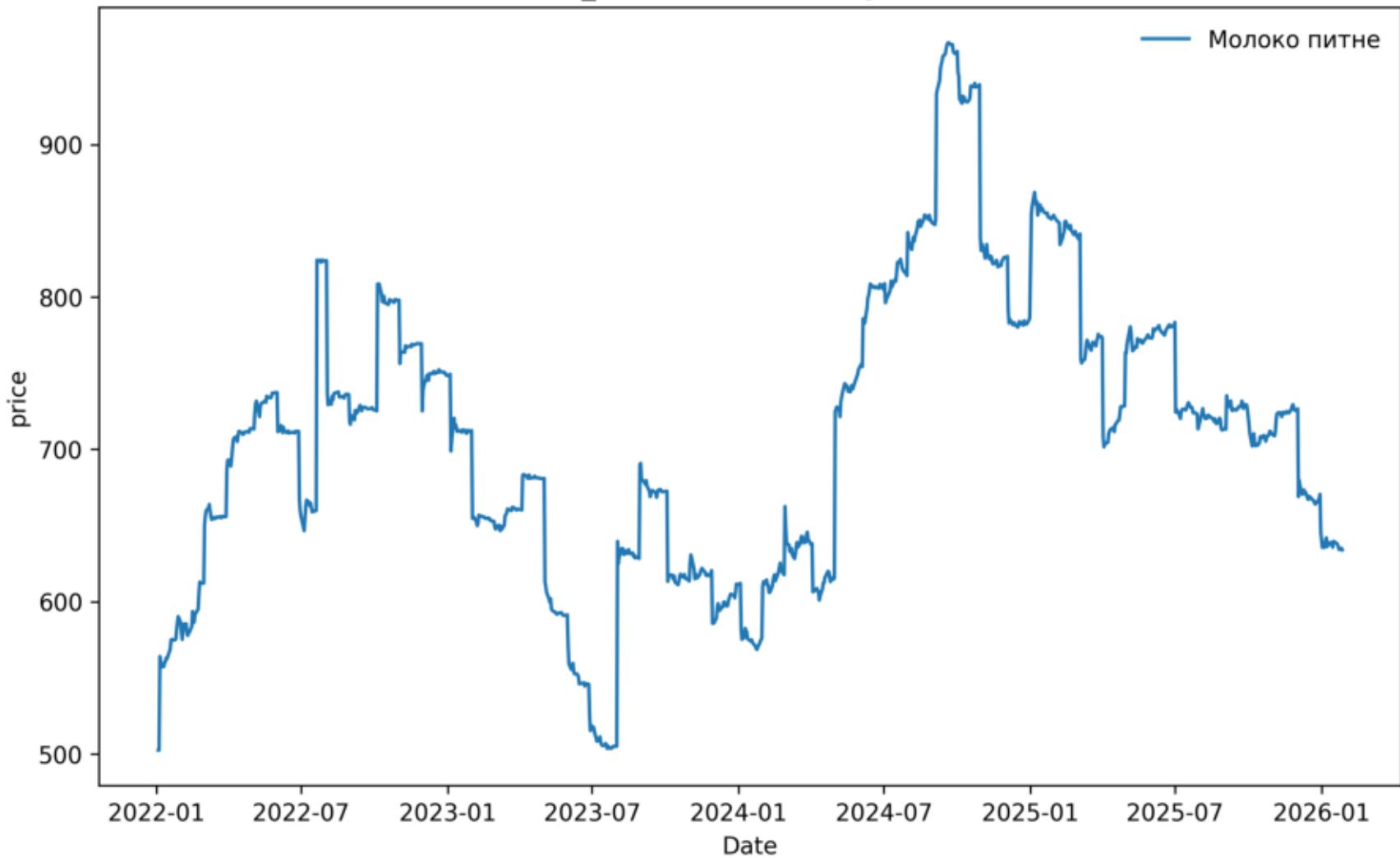
Table: descriptive_stats

source	product	standardized_type	series_variant	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30	
CME	Молоко	питне	milk	linear	0	1486	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	
CME	Молоко	питне	milk	pchip	0	1486	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	
CME	Молоко	питне	milk	real	1023	463	nan	709.2641239530792	713.331402	95.79149583199074	502.464444	966.9001849999999	564.4273536	635.788137	770.375485	854.4147727999999	0.2832414598568212	-0.07074089286782348	0.13505757953482467	0.010526815526692887	0.015164103628078825

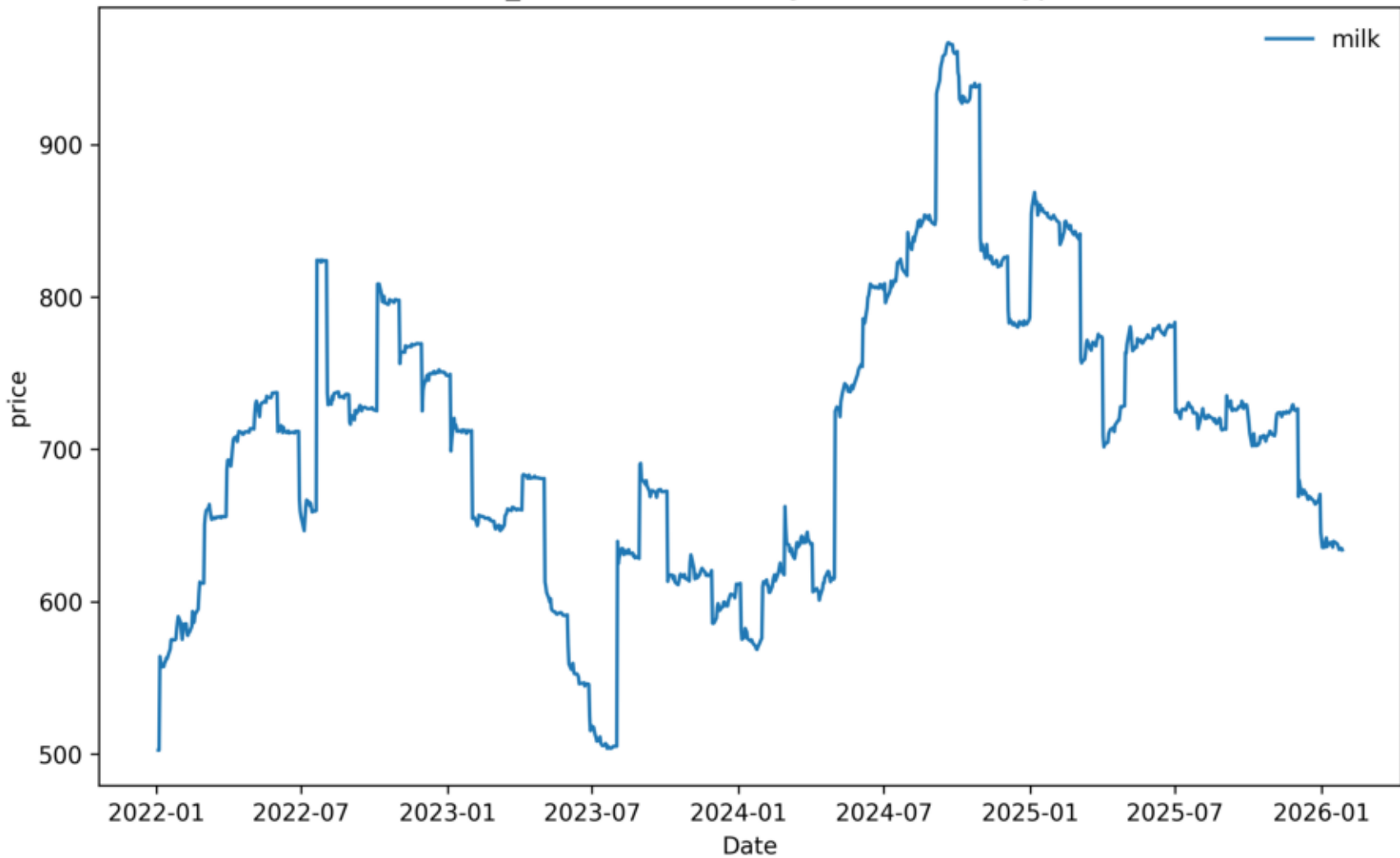
Table: tests

source	product	standardized_type	series_variant	n_obs	adf_p	kpss_p	ljungbox_p	bp_p	white_p	jb_p	stability_flag	stability_drift	action_label	integration_class	ac_risk	het_risk	non_normal_risk	stability_risk_class	recommended_action	recommended_model_family	avoid	diag_note
CME Молоко питне		milk	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated level OLS und...	Follow diagnostics-driven...
CME Молоко питне		milk	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated level OLS und...	Follow diagnostics-driven...
CME Молоко питне		milk	real	214	0.19809490728672302	0.044255554737156276	0.00233114991944491672	0.5837986248407785	0.647495693005676	8.722972988440486e-233	1	91.95170941168215	non-stationary: differenc...	I(1)-like	high	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated level OLS und...	Follow diagnostics-driven...

sheet_cme: Time Series by Product



sheet_cme: Time Series by Standardized Type



sheet_cme_distribution.png

sheet_cme: Distribution of price

